Exploratory Data Analysis (EDA) Report

Prepared based on the provided Python EDA script.

1. Objective

The script performs Exploratory Data Analysis on a student performance dataset. It loads the cleaned dataset, inspects its structure, summarizes statistics, checks data quality, and generates visualizations to understand score distributions and relationships.

2. Data Loading

The dataset is loaded from 'cleaned_student_data.csv' using pandas. The first five rows are displayed to verify successful loading and inspect sample records.

3. Dataset Structure Analysis

The info() function displays column names, data types, non-null counts, and memory usage. This helps identify numeric and categorical variables and detect potential data quality issues.

4. Statistical Summary

The describe() function provides count, mean, standard deviation, minimum, maximum, and quartile values for numerical columns. These metrics help understand central tendency and variation in student scores.

5. Missing Value Analysis

The script checks for missing values in every column. A dataset with zero missing values is generally considered ready for further analysis and machine learning tasks.

6. Correlation Analysis

A correlation matrix is generated for numerical columns. Correlation values range from -1 to +1. Positive values indicate direct relationships, negative values indicate inverse relationships, and values near zero indicate weak relationships.

7. Visualizations Generated

Histogram: Shows the distribution of MathScore values.
Bar Chart: Compares average MathScore and ScienceScore.
Box Plot: Detects score spread and potential outliers.
Heatmap: Visual representation of correlations among numerical features.

8. Output Files

The script saves the following images:
- Histogram.png
- BarChart.png
- BoxPlot.png
- Correlation_Heatmap.png

9. Key Insights Expected

The analysis helps identify score distributions, average academic performance, presence of outliers, missing data issues, and relationships between subject scores.

10. Conclusion

This EDA workflow provides a strong foundation for understanding student performance data before advanced analytics or machine learning. The generated visualizations and statistical summaries support data-driven decision making.